

Set-Theoretic Estimation over Non-Metric Domains
From Feasibility in Hilbert Space to Coherence in Language:
a Machine-Checked Monograph

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Abstract

Set-theoretic estimation (STE) answers an estimation problem by intersection: an estimate is acceptable exactly when it is consistent with every piece of available information. The classical theory [15] lives in metric — indeed Hilbert — space, where convexity and projection operators make the feasible set both meaningful and computable. This monograph is about what happens to STE *when the underlying set is non-metric*. We follow Carlson [10] in the topological origination of STE — set estimation over topological spaces with no metric, motivated by decryption — and then carry the theory into a domain where no metric was ever on offer: language. The carriage proceeds in three moves: first, a link to *parsing*, where Constraint Grammar’s eliminative algebra is shown to be exactly STE; second, *frame analysis*, where set-valued and dynamic frames give a possible-worlds semantics for machine-gathered meaning; third, accepting the challenge of *multi-document coreference*, where identity itself becomes a lattice limit — the least partition adequate to the accumulated evidence. What replaces the metric is order: the feasibility lattice, its sheaf-theoretic gluing, and a combinatorial Shannon geometry on readings. The monograph’s unifying thread is a single machine-checked dichotomy in that geometry: joining information is a metric contraction, extracting common information is not — and this join/meet conditioning asymmetry recurs, under three other names, as the tractable/hard, decomposed/coupled, and glueable/obstructed divides that organize every chapter. Every result cited as a theorem is verified in the project’s Lean 4 library and cited by exact identifier; everything else is framing, and is marked as such.

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Preface: how to read this book

This monograph is a *spine*, not an omnibus. The project it unifies has produced a body of papers and notes, each of which remains the owning, self-contained deep dive for its topic; this book supplies the connective narrative that makes them chapters of one argument, and it originates in full only the material that exists nowhere else in prose — chiefly the multi-document coreference theory of Chapter 10. Each part opens with synthesis that is new here, states what its topic contributes to the whole, and then delegates: “the full treatment is [paper].” Appendix A gives the complete ownership map, together with recommendations for trimming the residual overlaps among the existing papers.

Three disciplines hold throughout.

First, the mechanization bright line. Every claim labeled a theorem is machine-checked in the project’s Lean 4 / Mathlib [21, 55] library, sorry-free, verified in continuous integration, and cited by its exact Lean identifier, e.g. `frame_hasHelly_two (Ste.HellyConsistency)`. Surrounding prose is interpretation — ours to defend, never the kernel’s.

Second, coherence, not truth. The theory studies when a body of machine-gathered readings is *coherent* — jointly satisfiable — not whether it is *true*. The epistemological position, its bracketing of rival pictures of meaning, the fidelity/consistency conditional (a feasible set is only as good as the extraction that produced its constraints), and the Shannon-entropy ceiling on what crisp set-valued methods can express are stated once, carefully, in the project’s epistemology note (*Hyperframes and the Epistemology of Machine-Gathered Semantics*, `hyperframes-epistemology.tex`). We reference that position throughout and do not re-litigate it.

Third, no duplicated deep dives. Where a chapter’s material has an owning paper, this book states the result, its place in the arc, and its Lean anchor — and stops.

Part I

From Metric to Topological: the Classical Inheritance

Chapter 1

Estimation by intersection: the metric theory

Set-theoretic estimation begins with a reversal of habit. Classical estimation optimizes: it posits a criterion and returns the single point that extremizes it. STE instead *intersects*: every piece of information — a measurement, a bound, a prior structural fact — is recorded as the set of hypotheses consistent with it (its *property set*), and the deliverable is the *feasibility set*, the intersection of all property sets. Any member is an acceptable estimate; the set itself is the honest answer. Combettes [15] founded the framework in this generality, drawing on a lineage running through unknown-but-bounded state estimation [49], image recovery by convex projections [16, 58], and set-membership identification in signal processing [24].

The classical theory is emphatically *metric*. Its home is a Hilbert space; its property sets are closed and convex; its engine is the projection operator, well-defined precisely because a nearest point exists and is unique in that geometry. The great algorithmic corpus — POCS, parallel projection methods, and their convergence theory [4, 17] — is a corpus about metric structure: distances to sets decrease, Fejér monotonicity holds, fixed points attract. When textbooks say “set-theoretic estimation works,” they mostly mean: *the metric makes it work*.

This monograph asks what survives when the metric is removed — and the honest first answer is: the *definitions* survive untouched. Feasibility is an intersection; intersections need no metric. The consistent/fair/ideal taxonomy of estimators, information monotonicity (more constraints, smaller feasible set), and the detection of invalid information by inconsistency are order-theoretic facts. The project’s machine-checked core formalizes exactly this metric-free skeleton — the feasibility set and its lattice laws, the taxonomy, antitone refinement — and verifies that the classical statements hold with no appeal to distance. **The full treatment is the mechanization paper** (*A Machine-Checked Core for Set Theoretic Estimation*, `ste-mechanization.tex`), which owns the formal model and the core theorem list; the primary-source assessment of Combettes [15] is owned by the literature review (`literature-review.tex`).

What does *not* survive metric removal is the algorithmic:

no projections, no nearest points, no convergence-by-contraction. The burden of the rest of this book is that something else takes their place — order, lattice, and sheaf structure — and that the replacement is not a consolation prize but a theory with its own exact dichotomies, its own geometry, and its own frontier.

Chapter 2

The topological origination: Carlson

The first deliberate step off the metric was taken by Carlson [10], who recast STE over bare topological spaces to attack decryption. The move is conceptually prior to any application: if property sets are closed sets of a compact space, the feasibility set is closed, and nonemptiness — existence of an estimate — follows from the finite-intersection property, with no metric anywhere. Carlson’s cryptanalytic instantiation makes the stakes concrete: the hypothesis space is a key space, each intercepted ciphertext constrains it, and estimation is the shrinking of the feasible key set; the unicity phenomenon of Shannon [50] becomes a statement about when the feasibility set collapses to a point. A continuing line of work develops the program [7–9, 11–13].

For this monograph Carlson supplies three things. First, the *license*: STE over non-metric spaces is not our invention but an established origination that we follow and mechanize — the project’s library formalizes the closedness/compactness existence theorems, the skeleton of Carlson’s STE-decryption reduction, and his key-counting and cipher-reduction chain (owned by `ste-mechanization.tex`). Second, the *partition insight*: Carlson’s asymptotic equipartition work partition is STE-invariant — an early sign that when the metric goes, *partitions* (and their lattice) become the load-bearing structure; this observation is developed in the language note (`ste-and-language.tex`) via rough-set indiscernibility [46]. Third, the *template for a domain carriage*: Carlson carried STE from signal recovery to cryptanalysis by identifying the domain’s native constraint structure. Parts III–V of this book perform the same carriage into language.

The surveys own the surrounding scholarship: the projection-algorithm inheritance and the set-membership tradition (`topological-ste-survey.tex`), and the post-2012 uptake of topological STE together with the mechanization gap that this project fills (`decade-followon-survey.tex`). **Those two surveys, plus the literature review, are the owning deep dives for everything historical in this part**; we do not repeat their bibliographies here.

One more inheritance deserves its own flag, because it becomes the spine of the book. Shannon himself, in a paper that predates STE by four decades, described *the lattice theory of information* [51]: information sources ordered by

refinement, with join and meet. Modern work [25] equips that lattice with a metric derived from conditional entropy. When STE goes non-metric, it lands — provably, in the finite counting shadow — on exactly Shannon’s lattice. That landing is Chapter 3.

Part II

The Non-Metric Structure Theory

Chapter 3

The conditioning dichotomy: the spine of the book

This chapter states, once, the single mathematical fact that organizes the entire monograph. Later chapters will not restate it; they will point here.

3.1 The lattice that replaces the metric space

Remove the metric from STE and ask what structure the readings of a corpus actually carry. The project’s answer, accumulated across its Lean modules, is: a *partition lattice with a counting geometry*. Hypotheses about a corpus induce partitions (which claims corefer, which readings agree); partitions form a complete lattice; and on that lattice there is a genuine metric — not assumed, but derived. Define the combinatorial Shannon distance as the counting shadow of $D(X, Y) = H(X \mid Y) + H(Y \mid X)$:

$$\text{shannonDist}(P, Q) = (r(P \sqcup Q) - r(P)) + (r(P \sqcup Q) - r(Q)),$$

where r is the partition rank (`shannonDist`, `Ste.InfoDistance`; `rank`, `Ste.PartitionRank`). The triangle inequality is a theorem, and its proof is the submodularity of rank (`shannonDist_triangle`, `Ste.InfoDistance`; `rank_submodular`, `Ste.PartitionRank`) — the lattice-theoretic residue of the entropy inequalities, exactly as in the information-lattice synthesis of Delsol et al. [25] descending from Shannon [51].

The honest first observation kills a tempting misreading: on a finite carrier this metric is uniformly discrete — distinct partitions are at distance at least one (`one_le_shannonDist_of_ne`, `Ste.InfoTopology`) — so the induced point-set *topology* is trivial. The content of the non-metric theory is not topological but *metric-geometric*: it lies in how the lattice operations behave with respect to this distance.

3.2 The dichotomy

Here is the spine. The two lattice operations are *not* symmetric with respect to the Shannon geometry.

Theorem 3.1 (Join is a contraction). *For all partitions P, Q, R ,*

$$d(P \sqcup R, Q \sqcup R) \leq d(P, Q),$$

and join is jointly 1-Lipschitz: $d(P \sqcup Q, P' \sqcup Q') \leq d(P, P') + d(Q, Q')$. (`shannonDist_sup_right_le`, `shannonDist_sup_le_add`, `Ste.InfoTopology`.)

Theorem 3.2 (Meet is not — and has no uniform Lipschitz constant). *There exist partitions P, Q, R of a four-element carrier with $d(P, Q) < d(P \sqcap R, Q \sqcap R)$ (`shannonDist_inf_not_contraction`). More: for every constant C there exist P, Q, R — on the $2k$ -element carrier with $k = C + 1$, taking $P = \top$, Q the two halves, R the perfect matching — with $C \cdot d(P, Q) < d(P \sqcap R, Q \sqcap R)$, because $d(P, Q) = 1$ throughout the family while $d(P \sqcap R, Q \sqcap R) = k$ grows without bound (`shannonDist_meetWitness_eq`). The meet operation therefore admits no Lipschitz constant. (Both mechanized: `shannonDist_inf_not_contraction` and the general- k `meet_not_lipschitz`, `Ste.MeetScaling`.)*

Interpretation 3.3. Aggregating information — join, “all these readings together” — is well-conditioned: perturbing the inputs moves the aggregate by no more than the perturbation. *Extracting what readings share* — meet, common information — is ill-conditioned: an arbitrarily small change in the inputs can move the common part arbitrarily far. This is the counting shadow of the Gács–Körner phenomenon — common information is discontinuous and far less than mutual information [32] — surfacing inside STE with no probability in sight.

The remaining geometry sharpens the picture: the metric is bipartite-parity like a hypercube skeleton (`shannonDist_add_rank_add_rank`) yet embeds in no ℓ_1 cube (`pentagonal_violation`) and is not median (`median_not_unique`); it is geodesic along refinement chains (`aligned_of_chain`, `le_sup_of_aligned`) — the covering-graph distance of a geometric lattice, genuinely its own space (all `Ste.InfoTopology`). Coherence-space cannot be coordinatized away into a Hamming cube even in its crisp counting shadow.

3.3 The same dichotomy under three other names

monograph-level statement and is deliberately complete as such.

The claim that earns this chapter its title: the join/meet conditioning asymmetry is the *same* divide that the structure theory meets three more times, each occurrence machine-checked.

Tractable versus hard. For frame-structured corpora, pairwise consistency forces global consistency: the Helly number is two (`frame_hasHelly_two`, `Ste.HellyConsistency`). Checking coherence is a *join-side* activity — accumulate constraints, scan pairs — and it is correspondingly cheap and stable. The moment a question requires knowing what a family of readings has *in common* at higher order, locality fails: for second-order (set-of-sets) sources the Helly number provably jumps to at least three (`exists_not_hasHellySel_two`, `Ste.Carlson.SetOfSets`; Chapter 7).

Decomposed versus coupled. The partition rank is submodular (`rank_submodular`), and the *coupling excess* of a constraint against a product structure is zero exactly when the readings decompose and strictly positive exactly when they genuinely entangle (`couplingExcess_pos_of_not_le`, `Ste.CouplingRank`). Decomposable constraints are the ones whose information content is exhausted by joins of marginal information; coupling is precisely the residue that only a meet-like common-part analysis would see — and it is the exact source of representational blow-up (Chapter 5).

Gluable versus obstructed. Rectangular (product) constraints glue over every cover with vanishing Čech obstruction (`propertySet_cechVanishesCover`, `Ste.SetValuedFrame`); the diagonal — the archetypal coupled constraint — has obstruction exactly 2 (`diagonal_cechObstruction`, `Ste.CechObstruction`). Local-to-global success is the geometric face of decomposition; the obstruction is the geometric face of coupling (Chapter 4).

Interpretation 3.4. One dichotomy, four costumes: join/meet conditioning (well/ill-conditioned), Helly locality (tractable/hard), coupling excess (decomposed/coupled), Čech obstruction (gluable/obstructed). In each case the benign side is the side reachable by accumulating and combining — the join side — and the malignant side is the side that asks what parts share, agree on, or jointly determine — the meet side. Every subsequent chapter of this book instantiates the dichotomy in its own domain, and none of them escapes it.

The detailed geometry — proofs-in-prose, the non-embeddability witnesses, the asymptotic Rokhlin-metric outlook — is recorded in the interpretive log (`hyperframe-interpretations.tex`, the information-lattice section), which is the owning document for `Ste.InfoTopology` until its dedicated paper exists; this chapter is the

Chapter 4

Coherence as gluing: sheaves, obstructions, and the gluing complex

When the hypothesis space is a product over N variables, the feasibility set lives behind an exponential wall, and every practical scheme keeps only *local* information: small tables over small, overlapping sets of variables. Every such scheme lives or dies on one question — when does locally consistent partial information determine a global solution? The mathematics of that question is sheaf theory and Čech cohomology, transplanted to constraint satisfaction by the contextuality program of Abramsky and Brandenburger [1], Abramsky et al. [2], and developed for STE in machine-checked form by this project’s flagship paper.

The full treatment is the flagship (*Local Pieces, Global Solutions*, `ste-cohomology.tex`), whose centerpiece, the support–cover gluing theorem (`Ste.SupportCover`), characterizes exactly when every compatible family of local sections glues: precisely when the constraint is generated by pieces whose scopes fit inside the cover. Its accessible companion (`ste-cohomology-recast.tex`) re-derives the dictionary for readers with no topology; the transplant’s relation to the contextuality literature is owned by the obstruction lit-review (`cohomological-obstruction-litreview.tex`). Within the spine’s terms: gluing succeeds on the decomposed side of the dichotomy and is obstructed on the coupled side, with the diagonal’s obstruction of exactly 2 (`diagonal_cechObstruction`) as the minimal witness.

What this chapter adds — because it exists so far only in Lean and in the interpretive log — is the *refinement* of the gluing verdict into a shape. The module `Ste.LocalGluing` drops the all-or-nothing sheaf condition: `Glue sOn` asks whether a compatible family assembles over a *sub-cover*, and the family of sub-covers on which it does — the `gluingComplex` — is downward closed (`mem_gluingComplex_of_subset`), hence an abstract simplicial complex. The tight sheaf condition is exactly the presence of the top face (`cechVanishesCover_iff_gluesLocallyAt_card`), and the diagonal separates the notions: it glues locally at width one but not at width two (`diagonal_gluesLocallyAt_one`, `diagonal_not_gluesLocallyAt_two`, `diagonal_local_not_tight`). A bridge theorem returns to the spine’s tractable side: enough Helly structure on the

gluing fibers forces the top face — local coherence climbs to global (`cechVanishesCover_of_hasHelly_gluingFiber`).

Interpretation 4.1. Incoherence is not a single bit. A corpus that fails to cohere globally still coheres over a structured family of sub-corpora, and that family — the gluing complex — is the real object of study: the para-consistent stance [19] made geometric. The obstruction is thereby *located*, not merely detected. This becomes load-bearing for language in Part IV: a corpus of documents that contradict each other still has a shape of partial coherence, and Chapter 9’s ratification corpus exhibits exactly that shape in the wild.

The gluing complex is new material; its deep dive belongs to a future revision of the flagship (Appendix A records this assignment), and this section is its monograph-level statement.

Chapter 5

Representation and its limits

The gluing theory says *when* local pieces determine the whole; the representation theory says *what it costs*. The project's answer is a mechanized dichotomy stack: rectangular constraints are the tractable upper bound; coupling is the obstruction (`couplingExcess_pos_of_not_le`); block decomposition trades sums for products; conditioning and variable elimination are machine-checked; treewidth bounds the bags; and unbounded coupling is provably exponential — with the relation between obstruction size and representation blow-up a *sandwich, not an equality*. **The full treatment is the representation paper** (*Representation and Tractability in Set Theoretic Estimation*, `ste-representation-sheaves.tex`), with the sandwich's literature setting owned by `representation-bounds-litreview.tex`; the classical constraint-processing background is Dechter [22, 23], Robertson and Seymour [48], Yannakakis [56], and the lower-bound toolkit descends from communication complexity and extension complexity [31, 43].

In the spine's terms this chapter is the computational shadow of the dichotomy: everything join-like (accumulate, eliminate, decompose along a tree) is polynomial in the width; everything meet-like (what do these variables jointly determine that no projection sees) is where the exponential lives. We state this once and move on; the paper owns the proofs, and Appendix A flags its one overlap with the flagship for trimming.

Part III

Language I: Parsing

Chapter 6

Constraint Grammar is set-theoretic estimation

The first move into language is the most conservative one available: find the corner of computational linguistics that is *already* STE, and prove it. That corner is Constraint Grammar [40, 41]: begin with a universe of candidate morphosyntactic analyses, and let rules *eliminate* candidates until what survives is the parse. The practice matured through implemented parsers [5, 54], always described operationally, never estimation-theoretically.

The identification is now a theorem-level equivalence: for the hard, purely eliminative fragment, a CG rule *is* an STE property set, the surviving analyses *are* the feasibility set, rule addition is antitone refinement, fairness is preservation of the intended parse, and unambiguous successful analysis is ideal information — verified as an explicit equivalence in `Ste.ConstraintGrammar`. **The full treatment is the parsing note** (*The Algebraic Core of Constraint Grammar as Set Theoretic Estimation*, `constraint-grammar-ste.tex`), which also owns the scoping: CG operations that genuinely change the candidate universe are not isomorphic to static STE and must be treated as staged maps between STE problems. The broader historical claim — that unification grammar [52] and underspecification semantics [6, 18, 29] are likewise doing uncited STE, and that the bridge to Combettes [15] is unclaimed — is owned by the language note (`ste-and-language.tex`), together with the Formal Concept Analysis reading [34] in which feasible sets are extents of a concept lattice.

For the monograph’s arc, parsing contributes two lessons. First, *proof of carriage*: language admits STE not by analogy but by isomorphism, in at least one exactly-delimited fragment — the non-metric theory has a native application, with the domain supplying eliminative constraints where cryptanalysis supplied ciphertexts. Second, *the boundary lesson*: the isomorphism stops exactly where the candidate universe stops being fixed. A parser’s universe is one sentence’s analyses; a corpus’s universe of *readings* is neither fixed nor crisp. Pushing past that boundary is what forces the set-valued frame — the primitive of Part IV.

Part IV

Language II: Frames

Chapter 7

The set-valued frame and its deformations

The primitive of the language-side theory is not the crisp assertion “ $X = v$ ” but the *possibility set*: a source commits only to “the referent lies in S .” The epistemology note (`hyperframes-epistemology.tex`) owns the defense of this choice — reference as gathered by machines or readers is underdetermined but bounded, and pretending otherwise is false to the gathering — along with everything the position brackets. Here we record what the primitive *is*, machine-checked, and how it deforms.

The set-valued frame is a first-class object (`Ste.Set ValuedFrame`): each variable receives a possibility set; the induced constraint is the product box, hence rectangular (`propertySet.isRectangle`) and obstruction-free over every cover (`propertySet.cechVanishesCover`); a corpus is feasible exactly when every variable’s possibility-set intersection is nonempty (`corpusFeasibilitySet.nonempty_iff`); the feasibility set is a formal-concept extent (`isExtent.corpusFeasibilitySet`), unifying the possibility-set and concept-lattice [34] pictures; and the crisp determinate-reference model embeds as the singleton case with 2-locality transported (`hasHelly_ofCrisp.two`). In spine terms: the set-valued frame is engineered to live entirely on the benign side of the dichotomy — rectangular, gluable, Helly-2 — which is precisely why frame-based corpus checking is tractable at all.

Two deformations test how far the benign side extends.

Graded possibility. A fuzzy frame [59] grades membership in $[0, 1]$; feasibility becomes a `consistencyDegree`; and the α -cut theorem (`cut.fuzzyFeasibility`, `Ste.FuzzyFrame`) shows the graded theory is a *stack* of crisp theories indexed by confidence threshold — every crisp result acquires a graded shadow by cutting, the crisp theory sits at the boundary (`consistencyDegree.crisp_eq_one_iff`), and 2-locality survives in graded form (`weightedFrame.gradedHelly_two`). Grading is a *safe* deformation.

Higher-order possibility. A Carlson-style set-of-sets source — “the answer is one of these sets” — makes consistency a selectability problem (`SelectableConsistent`, `Ste.Carlson.SetOfSets`). The singleton case recovers the first-order theory exactly (`selectable_singleton_iff`, `hasHellySel_singletonFamily_iff`), but the deformation is *not* safe: a three-element triangle of two-element families is pairwise selectable with no global selection (`exists_not_hasHellySel_two`), so the second-order Helly

number is at least three. Disjunction over readings costs locality — the first recurrence of the spine’s dichotomy on language soil, and an operational correctness cliff: any solver ingesting set-of-sets data that relies on a pairwise scan is unsound.

The owning document for both deformations and their interpretation is the interpretive log (`hyperframe-interpretations.tex`), pending their dedicated paper; the epistemology note owns the position that the probabilistic deformation (a normalized density with Shannon entropy on top) is currently out of mechanized reach, and why the honest boundary is drawn there.

Chapter 8

Dynamic corpora: frames in motion

Real corpora are mutable: documents arrive, are deleted, and carry several candidate frame interpretations at once; coreference among their claims must not be forced to one early clustering. The project’s answer is a possible-worlds semantics: every exact interpretation and strict coreference partition is a candidate normalization hypothesis, constraints denote property sets of hypotheses, and the solver is literally STE. Identity statements then come in exact modal grades — inconsistent, must, cannot, uncertain — with must-coreference the safe quotient and the uncertainty overlay provably non-transitive. **The full treatment is the dynamic-frames paper** (*Dynamic Set-Theoretic Normalization*, `dynamic-frame-normalization.tex`): document-insertion antitonicity, deletion expansion, order independence, delta decomposition, minimal document supports, the 2^N tight limit, and an executable solver proved extensionally equal to the abstract semantics. The engineering face — how to shrink the feasible family while preserving the queries that matter, and where compression provably cannot go — is owned by the shrinking note (`ste-feasible-set-shrinking.tex`). The database lineage the paper builds on is conditional tables and provenance [35, 38], with truth-maintenance ancestry [20, 28].

One theorem from that development must be quoted here, because Chapter 10 is built directly on it. Must-coreference is the lattice meet of the feasible hypotheses’ partitions:

$$\text{mustSetoid}(D) = \text{sInf} \{ \text{exact partition of } h : h \in \text{feasible}(D) \}$$

(`mustSetoid_eq_sInf`, `Ste.PartitionRank`), monotone in the corpus (`mustSetoid_mono`, `rank_mustSetoid_mono`). This is Shannon’s lattice reading of the STE consensus: what *every* surviving hypothesis agrees on. Note its position in the spine: it is a *meet* over hypotheses — an agreement question — and the dichotomy warns that such questions are the ill-conditioned ones. The frontier chapter will take up the dual, join-side notion of identity and find it, by contrast, benign.

Chapter 9

Three corpora, three shapes

The theory has been run, end to end, on real text three times, with a fallible LLM extractor producing set-valued frames and the verified Lean pipeline doing everything from the frames onward. The three runs returned three different shapes of verdict, and together they span the gluing trichotomy of Chapter 4.

The tides corpus (multi-voice historical writing on tidal theory [26, 33, 42, 45, 47]): a consistent core exists after the machine *corrects* extraction slips — the inconsistency detector doing its founding job of invalid information detection. *The Federalist corpus* [36]: 86 documents pass through the proven pipeline and cohere at scale — the linear per-variable scan, proved complete for rectangular frame corpora, verifying feasibility cheaply exactly as the Helly-2 theory predicts. *The ratification two-camp corpus*: the consistent core *shatters*, cleanly and camp-wise — global gluing fails while camp-local coherence survives, the gluing complex exhibited in the wild. **The full treatments are the three results notes** (`ste-corpus-tides-results.tex`, `ste-corpus-federalist-results.tex`, `ste-corpus-ratification-twocamp.tex`), each of which states its own boundary-of-trust section; the fidelity/consistency conditional — the kernel verifies coherence of the extracted frames, never fidelity of extraction — is stated there and in the epistemology note, and we do not re-litigate it.

For the arc, the empirical chapters certify one thing: the non-metric theory is not a redescription but a running instrument, and its three possible verdicts — cohere, cohere-after-correction, shatter-with-located-obstruction — are exactly the three outcomes the structure theory said were possible.

Part V

Language III: the Frontier

Chapter 10

Multi-document coreference: identity as a lattice limit

This chapter is the monograph’s frontier and is originated here in full: its mathematics exists only in the Lean module `Ste.Coreference` and in the interpretive log, and this is its first complete prose treatment.

10.1 The problem, and why it is the right frontier

Everything in Part IV lived, formally, inside one mention space: the dynamic-frames development partitions a single `Claim` type, and its must-coreference is a single partition of it. But the questions that motivate corpus-scale semantics are cross-document by nature: *is the “General Carlson” of document 12 the “B. Carlson” of document 40?* Cross-document coreference resolution is a mature engineering problem [3, 14, 53], entangled with the hard ontology of event identity [37], and closely allied to record linkage and truth discovery in databases [27, 30, 44, 57]. The dominant method shape is the same everywhere: featurize pairs of mentions, score similarity, cluster — identity as the output of an optimizer.

The STE stance dictates a different decomposition of the problem, and it is the same decomposition the whole book has practiced: separate *what the evidence forces* from *what a scorer guesses*. A similarity model may propose links; that is extraction, fallible and outside the kernel. What the kernel can own is the question: *given* the per-document coreference decisions and a set of asserted cross-document links, what identifications follow? The answer is not a cluster assignment but a partition-lattice construction — and it turns out to have a universal property, exact monotonicity laws, a locality guardrail, and a place on the benign side of the spine’s dichotomy. Each of these is a machine-checked theorem.

10.2 Mentions with provenance

Fix a type `Doc` of documents and, for each d , a type `Local  $d$` of mention identifiers local to d . A *mention* is a dependent pair:

$$\text{Mention} = \Sigma_{d:\text{Doc}} \text{Local } d \quad (\text{Mention}, \text{Ste.Coreference}).$$

Provenance is thus carried in the type: two mentions from different documents are never conflated by construction, only ever by *evidence*. Each document contributes its local coreference decisions as a setoid (equivalence relation) `loc  $d$` on `Local  $d$` — the output of within-document resolution, which we take as given. The embedding `docSetoid` lifts `loc  $d$` to a setoid on all of `Mention` that relates mentions of document d as `loc  $d$` does and relates every other mention only to itself. The lift is provably provenance-respecting: it lies below the same-document kernel setoid `sameDocSetoid = ker( $\pi_1$ ) (docSetoid.le_sameDoc)`.

Cross-document evidence enters as a *raw relation* `Link  $\subseteq$  Mention  $\times$  Mention` — deliberately not assumed symmetric, transitive, or reflexive. This models the epistemic situation honestly: an extractor asserts individual links (“these two mentions co-refer”), and nothing about the assertion process guarantees closure properties. Equivalence is something we *impose* on evidence by closure, never something we *presume* of it.

10.3 The forced coreference and its universal property

Definition 10.1 (`forcedCoref`, `Ste.Coreference`). Given local coreferences `loc`, links `Link`, and an active document set $S \subseteq \text{Doc}$, the *corpus-forced coreference* is the join, in the complete lattice of setoids on `Mention`, of every active document’s contribution with the equivalence closure of the links:

$$\text{forcedCoref}(\text{loc}, \text{Link}, S) = \bigsqcup_{d \in S} \text{docSetoid}(\text{loc}, d) \sqcup \text{EqvGen}(\text{Link})$$

Call a setoid s *adequate* (to the evidence) if it dominates every active document’s contribution and contains every asserted link. The construction has the universal property one should demand of “what the evidence forces”:

Theorem 10.2 (Identity as a lattice limit; `forcedCoref_eq_sInf`). *The forced coreference is the least adequate setoid:*

$\text{forcedCoref}(\text{loc}, \text{Link}, S) = \text{sInf} \{ s \mid \forall d \in S, \text{docSetoid}(\text{loc}, d) \leq s \wedge \text{Link} \subseteq s \}$

Interpretation 10.3. “Who is the same as whom, across the corpus” is not a search problem whose answer must be guessed; it is a *fixed point of constraint accumulation* — the unique smallest partition consistent with all the local readings and all the asserted links. The join reading (close up the evidence) and the meet reading (intersect everything adequate) provably coincide, so the construction is canonical: any resolver that returns fewer identifications has ignored evidence, and any that returns more has invented some. Identity is *derived, not posited*.

The theorem is the exact dual of the single-document consensus theorem quoted in Chapter 8: there, `mustSetoid.eq.sInf` computes the *meet over hypotheses* — what every surviving world agrees on; here, `forcedCoref.eq.sInf` computes the *join over evidence* — the least commitment adequate to what was actually asserted. Both are `sInf` characterizations in the same lattice `Setoid`, indexed differently: by feasible worlds there, by document provenance here. This duality — consensus as meet, accumulation as join — is the spine’s dichotomy expressed inside a single application, and we return to it in §10.7.

10.4 Monotonicity: evidence only sharpens

Theorem 10.4 (`forcedCoref_mono_docs, forcedCoref_mono_link`). *The forced coreference is monotone in the active document set and in the link relation: $S \subseteq S'$ implies $\text{forcedCoref}(S) \leq \text{forcedCoref}(S')$, and pointwise $\text{Link} \subseteq \text{Link}'$ implies $\text{forcedCoref}(\text{Link}) \leq \text{forcedCoref}(\text{Link}')$.*

Theorem 10.5 (`rank_forcedCoref_mono_docs`). *On finite corpora the monotonicity is quantitative: adding documents can only raise the partition rank of the forced coreference, i.e. the number of merges the evidence has bought.*

Interpretation 10.6. Identity can only sharpen as documents arrive. This is the cross-document analogue of information monotonicity in classical STE — more constraints, smaller feasible set — but note the polarity flip that provenance indexing produces: in the dynamic-frames world, document insertion is *antitone* on the feasible hypothesis set; here it is *monotone* on forced identifications. Both statements are the same phenomenon (evidence accumulates) read on opposite sides of a Galois-style correspondence between hypotheses and commitments. Retraction — a document deleted, a link withdrawn — is deliberately outside this chapter’s formalism: the dynamic-frames paper owns deletion semantics, and grafting its deletion-expansion laws onto the provenance-indexed setting is future work (§10.8).

10.5 The locality guardrail

A resolver that can merge anything must be trusted not to merge everything. The formal system earns that trust with a separation theorem:

Theorem 10.7 (`forcedCoref_no_links_same_doc`). *With no cross-document links asserted, the forced coreference never crosses a document boundary: if $\text{forcedCoref}(\text{loc}, \emptyset, S)$ relates x and y , then x and y belong to the same document. (Via `forcedCoref_no_links_le_sameDoc`: the whole relation lies below the same-document kernel.)*

Interpretation 10.8. The formalism refuses to invent cross-document identity out of thin air. However rich each document’s internal coreference, and however many documents are active, no identification leaks across a boundary without a chain of asserted evidence. This is the guardrail that similarity-clustering architectures lack by construction: there, any two sufficiently similar mentions may be merged, and the burden of *not* merging namesakes falls on the score. Here the default is separation; merging is what must be paid for, link by link. In the epistemology note’s terms, the guardrail is part of the coherence discipline: the kernel polices what follows from evidence, and the fallibility of the links themselves is exactly where the fidelity/consistency conditional applies.

10.6 The witness: transitivity is the inference engine

The guardrail bounds the formalism from below; a concrete witness bounds it from above, showing the closure genuinely infers.

Theorem 10.9 (`witness_cross_doc_forced`). *Take three documents with one mention apiece, no internal coreference, and a chain of two links: document 0 to document 1, and document 1 to document 2 — but, provably, no direct 0–2 link (`chainLink_not_direct`), and the endpoints in genuinely different documents (`m0_ne_m2_doc`). The forced coreference identifies the document-0 mention with the document-2 mention.*

Interpretation 10.10. Together the guardrail and the witness bracket the system’s inferential power exactly: *nothing* without a chain of evidence, *everything* a chain licenses. All the inferential work is done by transitive closure over asserted links — which is also a precise statement of risk. One wrong link does not stay local: closure can propagate it into a cascade of merges. The monotonicity theorems then say the cascade is at least *orderly* (adding evidence never un-merges), and the rank corollary meters its size. A production resolver built on this kernel would therefore treat link assertion as the audited, expensive act — precisely the inversion of the similarity-clustering economy, where merging is cheap and auditing is impossible.

10.7 The spine, one last time

The forced coreference is a *join-side* construction: it accumulates evidence by $\sqcup$ in the setoid lattice. The spine’s contraction theorem (`shannonDist_sup_right_le`, Chapter 3) holds in every setoid lattice over a finite carrier, so it applies verbatim to the mention partitions of a finite corpus: perturbing one contribution — swapping a document’s local coreference for a nearby one, adding or dropping a link — moves the forced coreference by no more than the perturbation, in the combinatorial Shannon metric. Cross-document identity, as formalized here, is a *well-conditioned* quantity.

The meet-side questions are also present in this domain, and the dichotomy predicts their character. “What identifications do *all* feasible hypotheses agree on” (the `mustSetoid` consensus of Chapter 8) and “what do these two corpora’s coreference structures have in common” are $\sqcap$ -questions, and `shannonDist_inf_not_contract` warns that they are ill-conditioned: a small change in one input can move the agreement arbitrarily far. The practical reading is a design rule for corpus systems: *publish forced (join-side) identity; treat consensus (meet-side) identity as fragile, and never build downstream pipelines that differentiate through it*. That rule is this monograph’s thesis made operational, and it is the reason the frontier chapter belongs to the same book as the geometry chapter.

10.8 What is deliberately not yet mechanized

The chapter closes with its honest boundary, in the project’s standard form. Mechanized: everything cited above, sorry-free. Not mechanized, and flagged as the frontier’s frontier: (i) *link fallibility* — graded links with a fuzzy-frame lift (Chapter 7), so that a forced merge carries the minimum confidence along its evidence chain, in the spirit of subjective logic [39]; (ii) *retraction* — transporting the dynamic-frames deletion laws to the provenance-indexed setting; (iii) *entity-level frames* — attaching set-valued frames to merged clusters and asking when the merged frame remains feasible, which is where coreference re-enters the gluing theory of Chapter 4; (iv) *the Helly question* — whether 2-locality survives the sigma-type indexing of mentions, or whether provenance, like second-order possibility, has a locality price. Conjecture: it survives for link-free corpora (which are rectangular in the appropriate sense) and fails in general once links couple documents — that is, the dichotomy will reappear exactly here.

Chapter 11

The honest boundary

Every load-bearing caveat in this book has a single owning statement in the epistemology note (`hyperframes-epistemology.tex`), and this chapter is only its index. The theory studies *coherence*, *not truth*; its verdicts are conditional on extraction fidelity, and the kernel checks the consistency side of that conditional only. The crisp set-valued formalism has a stated expressiveness ceiling — no normalized densities, hence no genuine Shannon entropy over readings, until the mechanization landscape changes; the fuzzy stack of Chapter 7 is the possibilistic approximation, honestly labeled. The empirical chapters' boundary-of-trust sections instantiate the conditional per corpus. Nothing in this monograph strengthens, weakens, or renegotiates that position; a reader who wants it argued should read the note, which argues it at length.

Appendix A

The overlap-resolution map

This appendix is the monograph’s contract with its corpus: for every topic, exactly one document owns the deep dive; this book cites owners and never re-derives; and the residual overlaps among existing documents are flagged for trimming, as recommendations only — no existing paper is edited by this book.

A.1 Ownership

Topic	Owning deep dive
Formal STE core; taxonomy; Carlson counting/reduction chain	<code>ste-mechanization.tex</code>
Primary-source review (Combettes, Carlson)	<code>literature-review.tex</code>
Projection/convex inheritance; set-membership tradition	<code>topological-ste-survey.tex</code>
Post-2012 uptake; mechanization gap	<code>decade-followon-survey.tex</code>
Sheaf/Čech gluing for STE; support-cover theorem	<code>ste-cohomology.tex</code> (flagship)
Accessible gluing dictionary	<code>ste-cohomology-recast.tex</code>
Contextuality-literature setting of the obstruction	<code>cohomological-obstruction-litreview.tex</code>
Representation, treewidth stack, exponential lower bounds	<code>ste-representation-sheaves.tex</code>
Obstruction-vs-blow-up sandwich, literature	<code>representation-bounds-litreview.tex</code>
CG-as-STE equivalence and its scope	<code>constraint-grammar-ste.tex</code>
Historical bridge to CL; FCA/hyperframe lattice	<code>ste-and-language.tex</code>
Epistemology; coherence-not-truth; entropy ceiling	<code>hyperframes-epistemology.tex</code>
Dynamic corpus semantics; modal identity grades; 2^N limit	<code>dynamic-frame-normalization.tex</code>
Feasible-family reduction/compression contract	<code>ste-feasible-set-shrinking.tex</code>
New Lean threads (set-valued, fuzzy, set-of-sets, gluing complex, info-geometry): interpretations	<code>hyperframe-interpretations.tex</code> (running log)
Tides / Federalist / ratification runs	the three <code>ste-corpus-**-results</code> notes
Multi-document coreference (Ste.Coreference)	this monograph , Chapter 10
Conditioning dichotomy as unifying thread	this monograph , Chapter 3

A.2 Overlaps flagged for trimming (recommendations)

1. **Flagship vs. representation paper.** `ste-cohomology.tex` currently re-develops the bounded-treewidth algorithmic stack and the rectangle-cover / fooling-set bounds that `ste-representation-sheaves.tex` owns. Recommend: the flagship trims those sections to statements plus citations, keeping only what the support-cover theorem itself needs.
2. **Flagship vs. mechanization paper.** The flagship’s foundations sections (§2–3) re-explain the STE core and the Carlson turn owned by `ste-mechanization.tex` and the surveys. Recommend: compress to a one-page recap with pointers.
3. **Recast note vs. the metric-geometry thread.** `ste-cohomology-recast.tex`’s “readings form a metric space” section predates `Ste.InfoTopology` and now duplicates (in weaker form) material owned by the interpretive log and Chapter 3. Recommend: reduce to a pointer.
4. **The three surveys.** `literature-review.tex`, `topological-ste-survey.tex`, and `decade-follow-on-survey.tex` each re-summarize Carlson [10]’s core construction. Recommend: the two surveys cite the literature review’s summary instead of repeating it, keeping only their period-specific material.
5. **Dynamic frames vs. shrinking note.** The 2^N tight-limit theorem, minimal supports, and cross-corpus gluing appear in both `dynamic-frame-normalization.tex` and `ste-feasible-set-shrinking.tex`. Recommend: the shrinking note keeps the practitioner contract and cites the paper for the theorems.
6. **CG note vs. language note.** `constraint-grammar-ste.tex`’s “consequences transported” section partially re-derives monotonicity facts owned by `ste-and-language.tex`. Minor; recommend cross-reference.
7. **Tides plan vs. tides results.** `ste-corpus-tides-plan.tex` is superseded by the results note. Recommend: mark as historical/superseded in its header.
8. **Gluing complex placement.** `Ste.LocalGluing` currently has prose only in the log and in Chapter 4 here. Recommend: its deep dive be added to a future revision of the flagship (not to a new paper), keeping ownership consolidated.

Appendix B

Index of machine-checked results cited

All identifiers verified sorry-free in the project’s Lean 4 library at the time of writing; module names are relative to `lean/Ste/`.

Lean identifier	Module	Cited in
<code>frame_hasHelly_two</code>	<code>Ste.HellyConsistency</code>	Ch. 3
<code>rank_submodular</code>	<code>Ste.PartitionRank</code>	Ch. 3
<code>mustSetoid.eq.sInf</code>	<code>Ste.PartitionRank</code>	Ch. 8
<code>mustSetoid.mono</code>	<code>Ste.PartitionRank</code>	Ch. 8
<code>rank_mustSetoid.mono</code>	<code>Ste.PartitionRank</code>	Ch. 8
<code>couplingExcess_pos_of_not_le</code>	<code>Ste.CouplingRank</code>	Ch. 3
<code>shannonDist, shannonDist.triangle</code>	<code>Ste.InfoDistance</code>	Ch. 3
<code>one_le.shannonDist_of_ne</code>	<code>Ste.InfoTopology</code>	Ch. 3
<code>shannonDist_sup_right_le</code>	<code>Ste.InfoTopology</code>	Ch. 3, 10
<code>shannonDist_sup_le_add</code>	<code>Ste.InfoTopology</code>	Ch. 3
<code>shannonDist_inf_not_contraction</code>	<code>Ste.InfoTopology</code>	Ch. 3, 10
<code>shannonDist.add.rank.add.rank</code>	<code>Ste.InfoTopology</code>	Ch. 3
<code>pentagonal.violation, median_not_unique</code>	<code>Ste.InfoTopology</code>	Ch. 3
<code>aligned_of_chain, le_sup_of_aligned</code>	<code>Ste.InfoTopology</code>	Ch. 3
<code>diagonal_cechObstruction</code>	<code>Ste.CechObstruction</code>	Ch. 3, 4
<code>GluesOn, gluingComplex</code>	<code>Ste.LocalGluing</code>	Ch. 4
<code>mem_gluingComplex_of_subset</code>	<code>Ste.LocalGluing</code>	Ch. 4
<code>cechVanishesCover_iff_gluesLocallyAt_card</code>	<code>Ste.LocalGluing</code>	Ch. 4
<code>diagonal_gluesLocallyAt_one</code>	<code>Ste.LocalGluing</code>	Ch. 4
<code>diagonal_not_gluesLocallyAt_two</code>	<code>Ste.LocalGluing</code>	Ch. 4
<code>diagonal_local_not_right</code>	<code>Ste.LocalGluing</code>	Ch. 4
<code>cechVanishesCover_of_hasHelly_gluingFiber</code>	<code>Ste.LocalGluing</code>	Ch. 4
<code>propertySet_isRectangle</code>	<code>Ste.SetValuedFrame</code>	Ch. 7
<code>propertySet_cechVanishesCover</code>	<code>Ste.SetValuedFrame</code>	Ch. 3, 7
<code>corpusFeasibilitySet.nonempty_iff</code>	<code>Ste.SetValuedFrame</code>	Ch. 7
<code>isExtent.corpusFeasibilitySet</code>	<code>Ste.SetValuedFrame</code>	Ch. 7
<code>hasHelly_ofCrisp_two</code>	<code>Ste.SetValuedFrame</code>	Ch. 7
<code>consistencyDegree</code>	<code>Ste.FuzzyFrame</code>	Ch. 7
<code>cut_fuzzyFeasibility</code>	<code>Ste.FuzzyFrame</code>	Ch. 7
<code>consistencyDegree_crisp.eq.one_iff</code>	<code>Ste.FuzzyFrame</code>	Ch. 7

Bibliography

- [1] Samson Abramsky and Adam Brandenburger. The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics*, 13:113036, 2011. doi: 10.1088/1367-2630/13/11/113036. arXiv:1102.0264.
- [2] Samson Abramsky, Shane Mansfield, and Rui Soares Barbosa. The cohomology of non-locality and contextuality. *Electronic Proceedings in Theoretical Computer Science*, 95:1–14, 2012. doi: 10.4204/EPTCS.95.1. Proc. 8th International Workshop on Quantum Physics and Logic (QPL 2011). arXiv:1111.3620.
- [3] Shany Barhom, Vered Shwartz, Alon Eirew, Michael Bugert, Nils Reimers, and Ido Dagan. Revisiting joint modeling of cross-document entity and event coreference resolution. *Computational Linguistics*, 47(3):575–605, 2021. doi: 10.1162/coli.a.00414.
- [4] Heinz H. Bauschke and Jonathan M. Borwein. On projection algorithms for solving convex feasibility problems. *SIAM Review*, 38(3):367–426, 1996. doi: 10.1137/S0036144593251710.
- [5] Eckhard Bick and Tino Didriksen. CG-3—beyond classical constraint grammar. In *Proceedings of the 20th Nordic Conference of Computational Linguistics*, pages 31–39, 2015. URL <https://aclanthology.org/W15-1807/>.
- [6] Johan Bos. Predicate logic unplugged. In Paul Dekker and Martin Stokhof, editors, *Proceedings of the Tenth Amsterdam Colloquium*, pages 133–143, Amsterdam, 1996. ILLC/Department of Philosophy, University of Amsterdam.
- [7] Albert Carlson, Richard B. Wells, and Robert E. Hiromoto. Using set theoretic estimation to implement shannon secrecy theory. In *Proc. 3rd IEEE Int. Conf. on Intelligent Data Acquisition and Advanced Computing Systems (IDAACS)*, pages 435–438, Sofia, Bulgaria, September 2005. The seminal source of the STE-for-cryptanalysis line, seven years before Carlson’s dissertation. IEEE Xplore document 4062169, <https://ieeexplore.ieee.org/document/4062169>.
- [8] Albert Carlson, Bhaskar Ghosh, Indira Kalyan Dutta, Shivam Khare, and Michael Totaro. Key space reduction using isomorphs. In *IEEE Int. Conf. on Information Technology, Electronics and Mobile Communication (IEMCON)*, 2021.
- [9] Albert Carlson, Ryan Carlson, and Hans C. Mumm. A survey of entropy failure as an attack surface: Artificial intelligence and modern cryptanalysis. Technical report, Center for Governance of Technology and Systems (GoTech), University of Maryland, 2026. Institutional technical report; [https://gotech.umd.edu/sites/default/files/2026-05/EntrokeyAI%20\(8\).pdf](https://gotech.umd.edu/sites/default/files/2026-05/EntrokeyAI%20(8).pdf).
- [10] Albert H. Carlson. *Set Theoretic Estimation Applied to the Information Content of Ciphers and Decryption*. Ph.d. dissertation, University of Idaho, Moscow, Idaho, USA, May 2012. Retrieved via Internet Archive snapshot (2024-07-13) of the author’s ResearchGate full text, publication 349064680.
- [11] Albert H. Carlson, Robert E. Hiromoto, and Richard B. Wells. Breaking block and product ciphers applied across byte boundaries. *International Journal of Computing*, 12(3):259–266, 2014. Journal version of the IEEE IDAACS 2011 paper.
- [12] Albert H. Carlson, Sarangi R. Mikkilineni, Michael W. Totaro, Robert E. Hiromoto, and Richard B. Wells. An introduction to local entropy and local unicity. In *IEEE Int. Symp. on Networks, Computers and Communications (ISNCC)*, 2022. doi: 10.1109/ISNCC55209.2022.9851755.
- [13] Albert H. Carlson, Sarangi R. Mikkilineni, Michael W. Totaro, Richard B. Wells, and Robert E. Hiromoto. Equivalence of product ciphers to substitution ciphers and their security implications. In *IEEE Int. Symp. on Networks, Computers and Communications (ISNCC)*, 2022. doi: 10.1109/ISNCC55209.2022.9851719.
- [14] Arie Cattani, Alon Eirew, Gabriel Stanovsky, Martin Josifoski, and Ido Dagan. Sequential cross-document coreference resolution. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021. URL <https://aclanthology.org/2021.emnlp-main.382/>.
- [15] Patrick L. Combettes. The foundations of set theoretic estimation. *Proceedings of the IEEE*, 81(2):182–208, 1993. doi: 10.1109/5.214546. Retrieved from the

author’s page, <https://pcombet.math.ncsu.edu/proc.pdf>.

- [16] Patrick L. Combettes. The convex feasibility problem in image recovery. *Advances in Imaging and Electron Physics*, 95:155–270, 1996. doi: 10.1016/S1076-5670(08)70157-5.
- [17] Patrick L. Combettes and Jean-Christophe Pesquet. Proximal splitting methods in signal processing. *Fixed-Point Algorithms for Inverse Problems in Science and Engineering, Springer Optimization and Its Applications*, 49:185–212, 2011. doi: 10.1007/978-1-4419-9569-8_10.
- [18] Ann Copestake, Dan Flickinger, Carl Pollard, and Ivan A. Sag. Minimal recursion semantics: An introduction. *Research on Language and Computation*, 3(2–3):281–332, 2005.
- [19] Newton C. A. da Costa. On the theory of inconsistent formal systems. *Notre Dame Journal of Formal Logic*, 15(4):497–510, 1974. doi: 10.1305/ndjfl/1093891487.
- [20] Johan de Kleer. An assumption-based truth maintenance system. *Artificial Intelligence*, 28(2):127–162, 1986. doi: 10.1016/0004-3702(86)90080-9.
- [21] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction—CADE 28*, pages 625–635, 2021. doi: 10.1007/978-3-030-79876-5_37.
- [22] Rina Dechter. Bucket elimination: A unifying framework for reasoning. *Artificial Intelligence*, 113(1–2): 41–85, 1999. doi: 10.1016/S0004-3702(99)00059-4.
- [23] Rina Dechter. *Constraint Processing*. Morgan Kaufmann, San Francisco, 2003.
- [24] John R. Deller. Set membership identification in digital signal processing. *IEEE ASSP Magazine*, 6(4):4–20, 1989. doi: 10.1109/53.41661.
- [25] Idris Delsol, Olivier Rioul, Julien Béguinot, Victor Rabiet, and Antoine Souloumiac. An information theoretic condition for perfect reconstruction. *Entropy*, 26(1):86, 2024. doi: 10.3390/e26010086. PMC10814784. Shannon’s 1953 lattice of information with the Shannon and Rajske entropic metrics.
- [26] René Descartes. *Principia Philosophiae*. 1644. Part IV: tides as the pressure of the celestial vortex.
- [27] Xin Luna Dong, Laure Berti-Equille, and Divesh Srivastava. Data fusion: Resolving conflicts from multiple sources. In *Proceedings of the 35th International Conference on Very Large Data Bases*, 2009. URL <https://research.google/pubs/data-fusion-resolving-conflicts-from-multiple-sources/>.
- [28] Jon Doyle. A truth maintenance system. *Artificial Intelligence*, 12(3):231–272, 1979. doi: 10.1016/0004-3702(79)90008-0.
- [29] Markus Egg, Alexander Koller, and Joachim Niehren. The constraint language for lambda structures. *Journal of Logic, Language and Information*, 10(4):457–485, 2001.
- [30] Ahmed K. Elmagarmid, Panagiotis G. Ipeirotis, and Vassilios S. Verykios. Duplicate record detection: A survey. *IEEE Transactions on Knowledge and Data Engineering*, 19(1):1–16, 2007. doi: 10.1109/TKDE.2007.250581.
- [31] Samuel Fiorini, Serge Massar, Sebastian Pokutta, Hans Raj Tiwary, and Ronald de Wolf. Exponential lower bounds for polytopes in combinatorial optimization. *Journal of the ACM*, 62(2):17:1–17:23, 2015. doi: 10.1145/2716307. arXiv:1111.0837.
- [32] Peter Gács and János Körner. Common information is far less than mutual information. *Problems of Control and Information Theory*, 2(2):149–162, 1973.
- [33] Galileo Galilei. *Dialogue Concerning the Two Chief World Systems*. 1632. Fourth Day treats the tides; trans. Stillman Drake, Univ. of California Press, 1953.
- [34] Bernhard Ganter and Rudolf Wille. *Formal Concept Analysis: Mathematical Foundations*. Springer-Verlag, Berlin, Heidelberg, 1999. Translated by Cornelia Franzke.
- [35] Todd J. Green, Gregory Karvounarakis, and Val Tannen. Provenance semirings. In *Proceedings of PODS 2007*, pages 31–40, 2007. doi: 10.1145/1265530.1265535.
- [36] Alexander Hamilton, James Madison, and John Jay. *The Federalist Papers*. Independent Journal / New York Packet, 1788. 85 essays defending the proposed U.S. Constitution; public-domain text, Project Gutenberg eBook #18.
- [37] Eduard Hovy, Teruko Mitamura, Felisa Verdejo, Jun Araki, and Andrew Philpot. Events are not simple: Identity, non-identity, and quasi-identity. In *Workshop on Events: Definition, Detection, Coreference, and Representation*, pages 21–28, 2013. URL <https://aclanthology.org/W13-1203/>.
- [38] Tomasz Imieliński and Jr. Lipski, Witold. Incomplete information in relational databases. *Journal of the ACM*, 31(4):761–791, 1984. doi: 10.1145/1634.1886.
- [39] Audun Jøsang. *Subjective Logic: A Formalism for Reasoning Under Uncertainty*. Springer, Cham, 2016. doi: 10.1007/978-3-319-42337-1.

- [40] Fred Karlsson. Constraint grammar as a framework for parsing running text. In *COLING 1990 Volume 3: Papers Presented to the 13th International Conference on Computational Linguistics*, 1990. URL <https://aclanthology.org/C90-3030/>.
- [41] Fred Karlsson, Atro Voutilainen, Juha Heikkilä, and Arto Anttila, editors. *Constraint Grammar: A Language-Independent System for Parsing Unrestricted Text*. Number 4 in Natural Language Processing. Mouton de Gruyter, Berlin and New York, 1995. doi: 10.1515/9783110882629.
- [42] Johannes Kepler. *Astronomia Nova*. 1609. Introduction attributes the tides to the moon’s attraction.
- [43] Eyal Kushilevitz and Noam Nisan. *Communication Complexity*. Cambridge University Press, Cambridge, 1997. doi: 10.1017/CBO9780511574948. ISBN 978-0-521-56067-2.
- [44] Qi Li, Yaliang Li, Jing Gao, Bo Zhao, Wei Fan, Wei Han, and Jiawei Han. A survey on truth discovery. *SIGKDD Explorations Newsletter*, 17(2):1–16, 2015. doi: 10.1145/2897350.2897352. URL https://www.kdd.org/exploration_files/Article1_17_2.pdf.
- [45] Isaac Newton. *Philosophiæ Naturalis Principia Mathematica*. 1687. Book III: tides from the gravitation of sun and moon; trans. Andrew Motte, 1729.
- [46] Zdzisław Pawlak. Rough sets. *International Journal of Computer & Information Sciences*, 11(5):341–356, 1982.
- [47] Pliny the Elder. *Naturalis Historia*. 77. Book II records the correlation of tides with the moon and sun.
- [48] Neil Robertson and Paul D. Seymour. Graph minors. II. algorithmic aspects of tree-width. *Journal of Algorithms*, 7(3):309–322, 1986. doi: 10.1016/0196-6774(86)90023-4.
- [49] Fred C. Schweppe. Recursive state estimation: Unknown but bounded errors and system inputs. *IEEE Transactions on Automatic Control*, 13(1):22–28, 1968. doi: 10.1109/TAC.1968.1098790.
- [50] Claude E. Shannon. Communication theory of secrecy systems. *Bell System Technical Journal*, 28(4):656–715, 1949. doi: 10.1002/j.1538-7305.1949.tb00928.x.
- [51] Claude E. Shannon. The lattice theory of information. *Transactions of the IRE Professional Group on Information Theory*, 1(1):105–107, 1953.
- [52] Stuart M. Shieber. *An Introduction to Unification-Based Approaches to Grammar*. Number 4 in CSLI Lecture Notes. CSLI Publications, Stanford, CA, 1986.
- [53] Sameer Singh, Amarnag Subramanya, Fernando Pereira, and Andrew McCallum. Large-scale cross-document coreference using distributed inference and hierarchical models. In *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies*, pages 793–803, 2011. URL <https://research.google.com/pubs/archive/37560.pdf>.
- [54] Pasi Tapanainen and Atro Voutilainen. Tagging accurately—don’t guess if you know. In *Proceedings of the Fourth Conference on Applied Natural Language Processing*, pages 47–52, 1994. URL <https://aclanthology.org/A94-1008/>.
- [55] The Mathlib Community. *The Lean Mathematical Library*. 2020. doi: 10.1145/3372885.3373824.
- [56] Mihalis Yannakakis. Expressing combinatorial optimization problems by linear programs. *Journal of Computer and System Sciences*, 43(3):441–466, 1991. doi: 10.1016/0022-0000(91)90024-Y.
- [57] Xiaoxin Yin, Jiawei Han, and Philip S. Yu. Truth discovery with multiple conflicting information providers on the web. In *Proceedings of the 13th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 1048–1052, 2007. doi: 10.1145/1281192.1281301.
- [58] Dan C. Youla and Heywood Webb. Image restoration by the method of convex projections: Part 1—theory. *IEEE Transactions on Medical Imaging*, 1(2):81–94, 1982. doi: 10.1109/TMI.1982.4307555.
- [59] Lotfi A. Zadeh. Fuzzy sets. *Information and Control*, 8(3):338–353, 1965.